Supplementary Material: "Recovering Affective Signals in Baroque Opera: A Comparative Evaluation of Human and LLM Annotation"

This document is the supplementary material accompanying the paper "Recovering Affective Signals in Baroque Opera: A Comparative Evaluation of Human and LLM Annotation" by C. Vaquero Patricio and Á. Torrente, published in the Proceedings of the 27th International Society for Music Information Retrieval Conference (ISMIR), Abu Dhabi, UAE, 2026.

1 LLM Prompts

All LLM annotation was performed zero-shot using locally deployed models via Ollama, with temperature set to 0 and a fixed random seed (42) to ensure deterministic output. Each aria text (*prima parte* only) was presented to the model in a single-aria call. The system prompt identified the model role as: *"You are an expert emotion annotator in opera. Follow the instructions exactly."* The user prompt supplied the opera title, the aria text, and a numbered list of candidate labels, instructing the model to select exactly one emotion and return a structured JSON response:

```
OPERA: [opera title]
ARIA TEXT: [libretto text]
Choose EXACTLY ONE emotion from this numbered list:
1. [label_1] ... N. [label_N]
Reply with ONLY this JSON: {"id": "...", "emotion": "...", "confidence": 0.0}
```

For the **LLM-Proposed-SmallTax** conditions, a preliminary taxonomy generation call instructed the model — under the system prompt *"You generate emotion taxonomies for academic research"* — to produce N distinct English emotion words from a sample of ten aria texts, where N matched the cardinality of the reference human scheme and the six AriEmozione 1 categories were excluded to prevent collapse to a trivial vocabulary.
For the **LLM-DescartesTax** conditions, the model annotated all aria texts using the full Cartesian-based vocabulary of 92 descriptors. The resulting label distribution was inspected and labels with fewer than three supporting arias were excluded at evaluation time, following the same support threshold applied to all other conditions. Both Gemma and Phi-4 exhibit natural semantic convergence on this vocabulary, gravitating toward 7–8 categories.
For the **LLM-Proposed-LargeTax** conditions, the same agnostic induction procedure as LLM-Proposed-SmallTax was applied, with N matched to the number of surviving labels in the LLM-DescartesTax condition for the same model, enabling a controlled comparison between historically motivated and model-induced vocabularies at equivalent cardinality.

2 Label distributions

Label distributions for all 19 annotation conditions. $\mathbf{K_{raw}}$: total unique labels; $\mathbf{K_{filt}}$: labels surviving $m \geq 3$ aria texts; $\mathbf{N_{wo}}/\mathbf{N_{w}}$: compositions without/with outliers. Label counts show N aria texts. **Bold**: label survives filtering; plain text: excluded.

Family	Scheme	$\mathbf{K_{raw}}$	$\mathbf{K_{filt}}$	$\mathbf{N_{wo}}/\mathbf{N_{w}}$	Labels (N arias)
Human reference	AriEmozione 1	6	6	1437/1642	**sadness (23)**, **anger (22)**, **admiration (15)**, **love (10)**, **fear (9)**, **joy (5)**
	Annotator A	32	13	1013/1158	**pity (12)**, **anger (8)**, **boldness (4)**, **contempt (4)**, **fear (5)**, **sadness (5)**, **esteem (4)**, **love (3)**, **hope (4)**, **irresolution (3)**, **despair (3)**, jealousy (2), generosity (2), **abjectness (3)**, pride (2), **repentance (3)**, pride/abjectness (2), contempt/aversion (1), pride/boldness (1), repulsion (1), envy (1), distaste (1), fear/love (1), devotion (1), shame (1), despair/anger (1), disdain (1), veneration/boldness (1), glory (1), regret (1), desire/abjectness (1), contempt/pity (1)
	Annotator B	40	11	882/1007	**anger (7)**, **pity (5)**, **love (5)**, **sadness (8)**, **irresolution (6)**, **despair (4)**, **boldness (3)**, **confidence (3)**, **generosity (3)**, **fear (3)**, **despair/sadness (3)**, horror (1), hatred/boldness (1), pride (1), love/sadness (1), love/jealousy (1), pride/boldness (1), indignation/sadness (1), hope (2), sadness/shame (1), love/contempt/sadness (1), jealousy (1), admiration (2), shame,/joy (1), gratitude (1), hatred/pity (1), admiration/love (1), hatred (1), self-satisfaction (1), love/desire (1), fear/hope (1), humility (1), indignation/hatred (1), hope/sadness (1), repentance (1), desire (1), disdain/jealousy (1), contempt (1), joy (1), sadness/love (1)
LLM-HumanTax (Phi-4)	AriEm1 × Phi-4	6	5	1409/1610	**sadness (32)**, **fear (16)**, **love (13)**, **anger (14)**, **admiration (7)**, joy (2)
	AnnA × Phi-4	21	9	1127/1289	**boldness (14)**, **despair (13)**, **love (11)**, **pride/boldness (8)**, **contempt/aversion (5)**, **contempt (5)**, **disdain (3)**, **fear (3)**, pity (2), anger (2), **contempt/pity (3)**, shame (2), irresolution (2), fear/love (2), sadness (2), despair/anger (1), hope (2), jealousy (1), envy (1), generosity (1), regret (1)
	AnnB × Phi-4	28	9	1062/1215	**love/desire (15)**, **love/sadness (14)**, **despair/sadness (13)**, **indignation/sadness (5)**, **fear (4)**, **contempt (3)**, **sadness/love (3)**, pity (2), **hatred (3)**, indignation/hatred (2), **sadness (3)**, hatred/boldness (1), love (1), boldness (1), pride/boldness (1), disdain (1), love/jealousy (1), despair (1), sadness/shame (1), hope/love (1), self-satisfaction (1), irresolution (1), fear/hope (1), generosity (1), desire (1), admiration/love (1), anger (1), admiration (1)
LLM-HumanTax (Gemma)	AriEm1 × Gemma	6	5	1423/1626	**sadness (40)**, **anger (14)**, **admiration (13)**, **love (8)**, **fear (8)**, joy (1)
	AnnA × Gemma	16	9	1306/1493	**sadness (19)**, **despair (13)**, **pride (12)**, **devotion (9)**, **contempt/pity (7)**, **boldness (4)**, **love (4)**, **contempt/aversion (4)**, **hope (4)**, fear (2), pity (1), shame (1), remorse (1), irresolution (1), veneration/boldness (1), esteem (1)

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Family	Scheme	K_{raw}	K_{filt}	N_{wo}/N_w	Labels (N arias)
	AnnB × Gemma	27	7	1018/1167	**self-satisfaction (16)**, **sadness (13)**, **love/sadness (11)**, **indignation/sadness (10)**, **confidence (4)**, **admiration (4)**, **sadness/love (3)**, horror (2), pity (1), pride/boldness (1), hatred/boldness (2), love (1), hope (2), joy (1), disdain/jealousy (1), fear (1), indignation/hatred (1), desire (1), hatred/pity (1), repentance (1), hope/love (1), indignation (1), contempt (1), fear/hope (1), irresolution (1), love/desire (1), hope/sadness (1)
LLM-PSmall (Phi-4)	AriEm1 × Phi-4	6	6	1437/1642	**contempt (27)**, **anticipation (15)**, **surprise (11)**, **optimism (12)**, **trust (13)**, **disgust (6)**
	AnnA × Phi-4	21	10	1192/1363	**contempt (13)**, **remorse (9)**, **anxiety (10)**, **yearning (9)**, **pride (8)**, **jealousy (7)**, **hope (5)**, **nostalgia (3)**, **trust (3)**, **awe (3)**, guilt (2), optimism (2), confusion (2), excitement (1), shame (1), envy (1), calmness (1), surprise (1), interest (1), calm (1), satisfaction (1)
	AnnB × Phi-4	26	12	1184/1352	**satisfaction (8)**, **tenderness (9)**, **anxiety (8)**, **shame (7)**, **despair (7)**, **remorse (7)**, **grief (5)**, **melancholy (4)**, **trust (4)**, **hope (4)**, **jealousy (3)**, **awe (3)**, wonder (2), curiosity (1), excitement (1), guilt (1), panic (1), disgust (1), envy (1), resentment (1), calmness (1), hopelessness (1), inspiration (1), humor (1), pity (1), calm (1)
LLM-PSmall (Gemma)	AriEm1 × Gemma	6	6	1437/1642	**anxiety (20)**, **guilt (19)**, **hope (19)**, **disgust (14)**, **shame (8)**, **surprise (4)**
	AnnA × Gemma	19	12	1256/1436	**despair (17)**, **hope (9)**, **resentment (7)**, **trepidation (5)**, **longing (6)**, **sorrow (4)**, **frustration (6)**, **regret (5)**, **bewilderment (4)**, **tenderness (3)**, **contentment (4)**, dread (2), disgust (2), **unease (3)**, hatred (2), anxiety (2), remorse (1), envy (1), shame (1)
	AnnB × Gemma	26	10	1045/1195	**torment (12)**, **despair (8)**, **resentment (7)**, **sorrow (5)**, **yearning (7)**, **hope (6)**, **tenderness (4)**, **disappointment (5)**, **confusion (3)**, **regret (3)**, trepidation (2), remorse (2), longing (2), hatred (2), submission (2), peace (2), frustration (2), disgust (1), unease (1), worry (2), panic (1), envy (1), indignation (1), surprise (1), contentment (1), anxiety (1)
LLM-Descartes	Phi-4	15	8	1286/1470	**sadness/contempt (30)**, **admiration/love (11)**, **love/sadness (11)**, **boldness (7)**, **hope/confidence (6)**, **love/desire (4)**, **astonishment (3)**, **love/remorse (3)**, horror (2), disdain (2), remorse (1), hope/love (1), fear/hope (1), anger (1), anger/pity (1)
	Gemma	8	7	1416/1618	**sadness (34)**, **love/desire (13)**, **remorse (12)**, **hope/confidence (10)**, **fear (6)**, **astonishment (3)**, **irresolution (4)**, hope/sadness (2)
LLM-PLarge	Phi-4	14	7	1210/1382	**despair (27)**, **hope (12)**, **pride (8)**, **contempt (8)**, **anticipation (8)**, **guilt (4)**, gratitude (2), **trust (4)**, curiosity (2), envy (2), optimism (2), surprise (2), pessimism (2), shame (1)
	Gemma	8	7	1423/1626	**longing (23)**, **anxiety (14)**, **hope (14)**, **disgust (11)**, **shame (10)**, **guilt (6)**, **confusion (5)**, surprise (1)

2.1 Class imbalance

Class distributions for surviving labels ($m \geq 3$ aria texts) across all 19 annotation conditions (without outliers). Per-label counts show number of compositions. **Imb.**: majority/minority class ratio.

Family	Scheme	Imb.	Labels (N comps)
Human reference	AriEmozione 1	6.76	sadness (419), anger (401), admiration (247), love (160), fear (148), joy (62)
	Annotator A	6.12	pity (208), anger (147), boldness (83), contempt (83), fear (76), sadness (73), esteem (68), love (58), hope (49), irresolution (48), despair (45), abjectness (41), repentance (34)
	Annotator B	3.36	anger (141), pity (111), love (105), sadness (96), irresolution (89), despair (87), boldness (59), confidence (53), generosity (52), fear (47), despair/sadness (42)
LLM-HumanTax (Phi-4)	AriEm1 × Phi-4	4.77	sadness (520), fear (294), love (245), anger (241), admiration (109)
	AnnA × Phi-4	6.05	boldness (242), despair (230), love (219), pride/boldness (132), contempt/aversion (81), contempt (71), disdain (58), fear (54), contempt/pity (40)
	AnnB × Phi-4	7.19	love/desire (259), love/sadness (246), despair/sadness (202), indignation/sadness (89), fear (72), contempt (71), sadness/love (48), hatred (39), sadness (36)
LLM-HumanTax (Gemma)	AriEm1 × Gemma	4.81	sadness (664), anger (259), admiration (217), love (145), fear (138)
	AnnA × Gemma	6.46	sadness (310), despair (236), pride (223), devotion (158), contempt/pity (113), boldness (86), love (74), contempt/aversion (58), hope (48)
	AnnB × Gemma	5.87	self-satisfaction (264), sadness (228), love/sadness (186), indignation/sadness (150), confidence (79), admiration (66), sadness/love (45)
LLM-PSmall (Phi-4)	AriEm1 × Phi-4	4.34	contempt (438), anticipation (253), surprise (227), optimism (218), trust (200), disgust (101)
	AnnA × Phi-4	5.51	contempt (259), remorse (160), anxiety (152), yearning (141), pride (128), jealousy (121), hope (85), nostalgia (51), trust (48), awe (47)
	AnnB × Phi-4	3.22	satisfaction (164), tenderness (156), shame (128), anxiety (126), despair (124), remorse (102), grief (79), melancholy (73), trust (62), hope (60), jealousy (59), awe (51)
LLM-PSmall (Gemma)	AriEm1 × Gemma	4.32	anxiety (341), guilt (321), hope (296), disgust (236), shame (164), surprise (79)
	AnnA × Gemma	8.44	despair (287), hope (145), resentment (135), trepidation (101), longing (96), sorrow (96), frustration (89), regret (88), bewilderment (71), tenderness (65), contentment (49), unease (34)
	AnnB × Gemma	3.92	torment (188), despair (150), resentment (134), sorrow (123), yearning (116), hope (90), tenderness (75), disappointment (64), confusion (57), regret (48)
LLM-Descartes	Phi-4	13.62	sadness/contempt (545), admiration/love (190), love/sadness (176), boldness (130), hope/confidence (83), love/desire (74), astonishment (48), love/remorse (40)
	Gemma	8.91	sadness (588), love/desire (250), remorse (187), hope/confidence (146), fear (110), astonishment (69), irresolution (66)
LLM-PLarge	Phi-4	10.22	despair (470), hope (190), pride (156), contempt (140), anticipation (131), guilt (77), trust (46)
	Gemma	4.69	longing (417), anxiety (246), disgust (193), hope (192), shame (192), guilt (94), confusion (89)

3 Modeling results

				Without outliers			With outliers				
Family	**Condition**	**K**	**Imb.**	**Skill**	**95% CI**	p	**Skill**	**95% CI**	p	**Δ**	**Sil.**
Human	AriEmozione 1	6	6.76	**0.045**	[0.025, 0.068]	0.003	0.023	[−0.000, 0.042]	0.042	+0.022	−0.077
	Annotator A	13	6.12	0.024	[0.009, 0.042]	0.005	0.036	[0.022, 0.049]	0.001	−0.012	−0.097
	Annotator B	11	3.36	0.036	[0.012, 0.064]	0.010	0.039	[0.020, 0.060]	0.005	−0.003	−0.121
LLM-HTax	AriEm1×Phi4 (HT)	5	4.77	0.034	[0.011, 0.058]	0.010	*0.005*	[−0.013, 0.022]	*0.278*	+0.029	−0.017
	AriEm1×Gemma (HT)	5	4.81	**0.112**	[0.089, 0.135]	0.001	0.098	[0.063, 0.127]	0.002	+0.014	−0.032
	AnnA×Phi4 (HT)	9	6.05	*0.019*	[−0.003, 0.042]	*0.065*	0.023	[0.004, 0.042]	0.032	−0.004	−0.107
	AnnA×Gemma (HT)	9	6.46	0.048	[0.038, 0.060]	0.001	0.048	[0.036, 0.059]	0.001	+0.001	−0.095
	AnnB×Phi4 (HT)	9	7.19	*0.014*	[−0.004, 0.031]	*0.097*	0.019	[0.010, 0.028]	0.005	−0.005	−0.053
	AnnB×Gemma (HT)	7	5.87	0.017	[0.003, 0.032]	0.032	*0.018*	[−0.002, 0.040]	*0.065*	−0.001	−0.079
LLM-PSmall	AriEm1×Phi4 (PS)	6	4.34	*0.017*	[0.001, 0.034]	*0.065*	0.030	[0.001, 0.057]	0.024	−0.014	−0.064
	AriEm1×Gemma (PS)	6	4.32	**0.064**	[0.039, 0.088]	0.002	0.061	[0.037, 0.085]	0.002	+0.003	−0.030
	AnnA×Phi4 (PS)	10	5.51	0.024	[0.009, 0.039]	0.014	0.014	[0.002, 0.025]	0.032	+0.011	−0.095
	AnnA×Gemma (PS)	12	8.44	0.017	[0.000, 0.033]	0.042	0.013	[0.004, 0.024]	0.014	+0.003	−0.078
	AnnB×Phi4 (PS)	12	3.22	0.050	[0.028, 0.073]	0.002	0.045	[0.018, 0.069]	0.007	+0.005	−0.114
	AnnB×Gemma (PS)	10	3.92	*0.015*	[-0.000, 0.032]	*0.116*	*0.005*	[−0.006, 0.017]	*0.348*	+0.010	−0.104
LLM-Desc	Phi4 (Desc)	8	13.62	**0.050**	[0.028, 0.073]	0.005	0.041	[0.023, 0.062]	0.001	+0.010	−0.137
	Gemma (Desc)	7	8.91	*0.008*	[−0.029, 0.043]	*0.348*	*0.009*	[−0.016, 0.033]	*0.246*	−0.001	−0.066
LLM-PLarge	Phi4 (PLarge)	7	10.22	**0.032**	[0.019, 0.044]	0.002	0.040	[0.021, 0.059]	0.003	−0.007	−0.054
	Gemma (PLarge)	7	4.68	0.031	[0.015, 0.047]	0.010	0.018	[0.005, 0.029]	0.024	+0.013	−0.049

Table 3: Full results (Logistic Regression, aria-grouped splits). **K**: label count after support filtering ($m \geq 3$ arias); **Imb.**: majority/minority class ratio; **Skill**: $(F1_{\text{model}} - F1_{\text{dummy}})/(1 - F1_{\text{dummy}})$; **Δ**: skill without − skill with outliers; **Sil.**: mean silhouette (without outliers). Italicised values: $p \geq 0.05$. Bold: best significant skill per family (without outliers).

3.1 Per-label F1 and skill scores grouped by annotation condition.

Table 4: Class distributions for surviving labels ($m \geq 3$ aria texts) across all 19 annotation conditions (without outliers).

Target	Label	Supp.	F1	95% CI	Skill	95% CI
AriEmozione 1	sadness	419	0.290	[0.231, 0.350]	−0.006	[-0.101, 0.083]
	anger	401	0.261	[0.193, 0.329]	+0.092	[0.009, 0.167]
	admiration	247	0.212	[0.156, 0.268]	+0.014	[-0.063, 0.089]
	love	160	0.204	[0.150, 0.258]	+0.125	[0.061, 0.187]
	fear	148	0.133	[0.075, 0.191]	+0.027	[-0.060, 0.103]
	joy	62	0.042	[0.006, 0.077]	+0.017	[-0.023, 0.055]
Annotator A	pity	208	0.226	[0.109, 0.343]	+0.057	[-0.047, 0.172]
	anger	147	0.143	[0.052, 0.233]	+0.035	[-0.006, 0.081]
	boldness	83	0.063	[0.000, 0.137]	+0.041	[-0.022, 0.107]
	contempt	83	0.015	[0.000, 0.041]	−0.044	[-0.119, 0.017]
	fear	76	0.045	[0.000, 0.098]	+0.008	[-0.034, 0.050]
	sadness	73	0.095	[0.021, 0.170]	+0.049	[-0.037, 0.128]
	esteem	68	0.035	[0.000, 0.070]	+0.024	[-0.008, 0.055]
	love	58	0.051	[0.000, 0.113]	+0.013	[-0.089, 0.101]
	hope	49	0.084	[0.038, 0.130]	+0.043	[-0.010, 0.092]
	irresolution	48	0.099	[0.014, 0.184]	+0.077	[0.002, 0.154]
	despair	45	0.042	[0.013, 0.072]	+0.015	[-0.008, 0.036]
	abjectness	41	0.077	[0.005, 0.150]	+0.054	[-0.014, 0.123]
	repentance	34	0.007	[0.000, 0.023]	−0.006	[-0.048, 0.032]
Annotator B	anger	141	0.170	[0.102, 0.238]	+0.004	[-0.057, 0.065]
	pity	111	0.019	[0.000, 0.043]	−0.069	[-0.160, 0.004]
	love	105	0.254	[0.116, 0.393]	+0.176	[0.023, 0.324]
	sadness	96	0.137	[0.035, 0.239]	+0.027	[-0.067, 0.121]
	irresolution	89	0.066	[0.028, 0.104]	−0.013	[-0.067, 0.036]
	despair	87	0.086	[0.020, 0.152]	+0.020	[-0.046, 0.082]
	boldness	59	0.040	[0.000, 0.130]	+0.011	[-0.084, 0.104]
	confidence	53	0.147	[0.031, 0.263]	+0.121	[0.027, 0.223]
	generosity	52	0.045	[0.001, 0.089]	+0.016	[-0.035, 0.064]
	fear	47	0.049	[0.000, 0.122]	+0.030	[-0.017, 0.082]
	despair,sadness	42	0.045	[0.000, 0.097]	+0.009	[-0.068, 0.077]
AriEm1 × Phi-4 (HTax)	sadness	520	0.257	[0.199, 0.315]	−0.119	[-0.239, -0.021]
	fear	294	0.231	[0.160, 0.302]	+0.053	[-0.017, 0.118]
	love	245	0.114	[0.064, 0.165]	−0.006	[-0.081, 0.061]
	anger	241	0.301	[0.249, 0.354]	+0.168	[0.100, 0.230]
	admiration	109	0.085	[0.014, 0.156]	+0.036	[-0.030, 0.103]
AnnA × Phi-4 (HTax)	boldness	242	0.169	[0.106, 0.232]	−0.056	[-0.121, 0.009]
	despair	230	0.150	[0.097, 0.203]	−0.081	[-0.181, 0.007]
	love	219	0.225	[0.123, 0.327]	+0.139	[0.042, 0.236]
	pride/boldness	132	0.134	[0.072, 0.195]	+0.029	[-0.024, 0.080]
	contempt/aversion	81	0.032	[0.000, 0.072]	−0.015	[-0.087, 0.048]
	contempt	71	0.066	[0.018, 0.113]	+0.010	[-0.071, 0.081]
	disdain	58	0.078	[0.021, 0.136]	+0.018	[-0.085, 0.104]
	fear	54	0.082	[0.004, 0.160]	+0.063	[-0.005, 0.132]
	contempt/pity	40	0.081	[0.030, 0.131]	+0.067	[0.023, 0.111]
AnnB × Phi-4 (HTax)	love/desire	259	0.129	[0.109, 0.150]	−0.137	[-0.216, -0.071]
	love/sadness	246	0.368	[0.268, 0.468]	+0.194	[0.122, 0.273]
	despair/sadness	202	0.178	[0.146, 0.210]	+0.022	[-0.053, 0.086]
	indignation/sadness	89	0.077	[0.027, 0.127]	−0.009	[-0.068, 0.041]

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Target	Label	Supp.	F1	95% CI	Skill	95% CI
	fear	72	0.096	[0.043, 0.148]	+0.052	[-0.010, 0.106]
	contempt	71	0.040	[0.000, 0.082]	+0.002	[-0.031, 0.033]
	sadness/love	48	0.000	–	−0.007	[-0.025, 0.010]
	hatred	39	0.014	[0.000, 0.045]	+0.014	[-0.017, 0.045]
	sadness	36	0.008	[0.000, 0.027]	−0.004	[-0.045, 0.033]
AriEm1 × Gemma (HTax)	sadness	664	0.441	[0.364, 0.517]	+0.019	[-0.065, 0.110]
	anger	259	0.262	[0.213, 0.311]	+0.113	[0.046, 0.173]
	admiration	217	0.263	[0.186, 0.340]	+0.124	[0.063, 0.187]
	love	145	0.116	[0.063, 0.169]	+0.033	[-0.007, 0.074]
	fear	138	0.279	[0.191, 0.367]	+0.200	[0.092, 0.304]
AnnA × Gemma (HTax)	sadness	310	0.224	[0.184, 0.264]	+0.008	[-0.079, 0.081]
	despair	236	0.185	[0.112, 0.258]	+0.010	[-0.045, 0.066]
	pride	223	0.130	[0.045, 0.215]	+0.017	[-0.028, 0.068]
	devotion	158	0.147	[0.102, 0.193]	+0.065	[0.008, 0.118]
	contempt/pity	113	0.146	[0.050, 0.242]	+0.079	[-0.052, 0.197]
	boldness	86	0.142	[0.019, 0.264]	+0.099	[-0.025, 0.224]
	love	74	0.096	[0.025, 0.168]	+0.056	[-0.002, 0.116]
	contempt/aversion	58	0.097	[0.044, 0.150]	+0.078	[0.023, 0.131]
	hope	48	0.032	[0.000, 0.063]	−0.005	[-0.062, 0.044]
AnnB × Gemma (HTax)	self-satisfaction	264	0.205	[0.138, 0.273]	−0.091	[-0.222, 0.008]
	sadness	228	0.115	[0.071, 0.159]	−0.048	[-0.092, -0.008]
	love/sadness	186	0.219	[0.165, 0.273]	+0.048	[-0.027, 0.119]
	indignation/sadness	150	0.119	[0.092, 0.147]	+0.013	[-0.028, 0.049]
	confidence	79	0.048	[0.004, 0.092]	+0.029	[-0.016, 0.074]
	admiration	66	0.078	[0.002, 0.154]	+0.040	[0.005, 0.079]
	sadness/love	45	0.102	[0.013, 0.190]	+0.091	[0.017, 0.170]
AriEm1 × Phi-4 (PSmall)	contempt	438	0.219	[0.189, 0.248]	−0.091	[-0.134, -0.052]
	anticipation	253	0.185	[0.120, 0.250]	+0.025	[-0.051, 0.093]
	surprise	227	0.083	[0.037, 0.129]	−0.059	[-0.115, -0.011]
	optimism	218	0.119	[0.088, 0.150]	−0.026	[-0.090, 0.030]
	trust	200	0.182	[0.106, 0.258]	+0.094	[0.030, 0.160]
	disgust	101	0.128	[0.064, 0.192]	+0.059	[-0.006, 0.122]
AnnA × Phi-4 (PSmall)	contempt	259	0.124	[0.049, 0.199]	−0.052	[-0.122, 0.016]
	remorse	160	0.177	[0.097, 0.256]	+0.056	[-0.028, 0.140]
	anxiety	152	0.067	[0.000, 0.137]	−0.006	[-0.039, 0.026]
	yearning	141	0.130	[0.055, 0.205]	+0.069	[-0.025, 0.156]
	pride	128	0.203	[0.131, 0.275]	+0.113	[0.050, 0.177]
	jealousy	121	0.094	[0.051, 0.137]	+0.032	[-0.020, 0.079]
	hope	85	0.082	[0.013, 0.150]	+0.036	[-0.024, 0.096]
	nostalgia	51	0.000	–	−0.014	[-0.052, 0.020]
	trust	48	0.039	[0.000, 0.078]	+0.034	[-0.007, 0.074]
	awe	47	0.035	[0.000, 0.090]	+0.028	[-0.032, 0.087]
AnnB × Phi-4 (PSmall)	satisfaction	164	0.206	[0.115, 0.298]	+0.105	[0.038, 0.179]
	tenderness	156	0.204	[0.129, 0.279]	+0.090	[0.029, 0.153]
	shame	128	0.084	[0.042, 0.126]	−0.002	[-0.037, 0.030]
	anxiety	126	0.295	[0.188, 0.402]	+0.205	[0.084, 0.323]
	despair	124	0.084	[0.050, 0.118]	+0.002	[-0.096, 0.082]
	remorse	102	0.080	[0.000, 0.173]	+0.055	[-0.042, 0.150]
	grief	79	0.054	[0.000, 0.112]	−0.006	[-0.080, 0.061]
	melancholy	73	0.128	[0.060, 0.197]	+0.085	[0.035, 0.138]
	trust	62	0.031	[0.001, 0.060]	−0.007	[-0.057, 0.039]

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Table 4 – Continued from previous page

Target	Label	Supp.	F1	95% CI	Skill	95% CI
	hope	60	0.039	[0.000, 0.086]	+0.021	[-0.021, 0.062]
	jealousy	59	0.063	[0.000, 0.138]	+0.053	[-0.025, 0.129]
	awe	51	0.042	[0.000, 0.108]	+0.015	[-0.037, 0.068]
AriEm1 × Gemma (PSmall)	anxiety	341	0.259	[0.209, 0.309]	+0.058	[-0.045, 0.146]
	guilt	321	0.288	[0.221, 0.354]	+0.079	[0.010, 0.150]
	hope	296	0.221	[0.188, 0.254]	+0.030	[-0.030, 0.085]
	disgust	236	0.248	[0.187, 0.309]	+0.119	[0.036, 0.195]
	shame	164	0.140	[0.086, 0.194]	+0.052	[-0.020, 0.118]
	surprise	79	0.097	[0.046, 0.149]	+0.064	[0.001, 0.123]
AnnA × Gemma (PSmall)	despair	287	0.124	[0.061, 0.188]	−0.097	[-0.206, -0.002]
	hope	145	0.151	[0.087, 0.215]	+0.101	[0.023, 0.174]
	resentment	135	0.121	[0.056, 0.187]	+0.044	[0.004, 0.085]
	trepidation	101	0.131	[0.047, 0.214]	+0.081	[0.012, 0.153]
	longing	96	0.057	[0.020, 0.093]	−0.022	[-0.086, 0.034]
	sorrow	96	0.040	[0.000, 0.089]	−0.007	[-0.071, 0.052]
	frustration	89	0.087	[0.035, 0.139]	−0.012	[-0.077, 0.047]
	regret	88	0.055	[0.015, 0.094]	−0.024	[-0.084, 0.029]
	bewilderment	71	0.088	[0.038, 0.138]	+0.042	[-0.004, 0.087]
	tenderness	65	0.003	[0.000, 0.010]	−0.066	[-0.136, -0.008]
	contentment	49	0.071	[0.015, 0.127]	+0.055	[0.013, 0.099]
	unease	34	0.000	–	+0.000	–
AnnB × Gemma (PSmall)	torment	188	0.110	[0.070, 0.150]	−0.064	[-0.142, 0.004]
	despair	150	0.073	[0.036, 0.110]	−0.098	[-0.152, -0.049]
	resentment	134	0.217	[0.112, 0.323]	+0.143	[0.054, 0.237]
	sorrow	123	0.032	[0.000, 0.064]	−0.064	[-0.133, -0.006]
	yearning	116	0.093	[0.035, 0.150]	+0.038	[0.006, 0.071]
	hope	90	0.089	[0.037, 0.140]	+0.018	[-0.078, 0.101]
	tenderness	75	0.008	[0.000, 0.020]	−0.038	[-0.087, 0.005]
	disappointment	64	0.131	[0.115, 0.146]	+0.099	[0.040, 0.149]
	confusion	57	0.055	[0.000, 0.113]	+0.026	[-0.038, 0.087]
	regret	48	0.007	[0.000, 0.023]	+0.007	[-0.009, 0.023]
Phi-4 (Descartes)	sadness/contempt	545	0.286	[0.249, 0.322]	−0.113	[-0.212, -0.038]
	admiration/love	190	0.171	[0.106, 0.235]	+0.041	[-0.005, 0.085]
	love/sadness	176	0.189	[0.113, 0.265]	+0.079	[-0.009, 0.162]
	boldness	130	0.207	[0.098, 0.317]	+0.161	[0.057, 0.268]
	hope/confidence	83	0.088	[0.032, 0.144]	+0.006	[-0.088, 0.089]
	love/desire	74	0.003	[0.000, 0.009]	−0.028	[-0.063, 0.003]
	astonishment	48	0.076	[0.036, 0.116]	+0.047	[0.008, 0.084]
	love/remorse	40	0.133	[0.000, 0.279]	+0.101	[-0.086, 0.265]
Gemma (Descartes)	sadness	588	0.257	[0.213, 0.301]	−0.222	[-0.358, -0.126]
	love/desire	250	0.151	[0.091, 0.211]	+0.035	[-0.046, 0.110]
	remorse	187	0.190	[0.127, 0.254]	+0.087	[0.033, 0.141]
	hope/confidence	146	0.130	[0.077, 0.183]	+0.046	[-0.017, 0.106]
	fear	110	0.117	[0.040, 0.193]	+0.067	[-0.014, 0.147]
	astonishment	69	0.042	[0.006, 0.077]	+0.009	[-0.020, 0.038]
	irresolution	66	0.069	[0.000, 0.139]	+0.045	[-0.019, 0.110]
Phi-4 (PLarge)	despair	470	0.321	[0.267, 0.374]	−0.165	[-0.230, -0.096]
	hope	190	0.214	[0.109, 0.318]	+0.118	[0.042, 0.203]
	pride	156	0.214	[0.101, 0.327]	+0.133	[0.046, 0.228]
	contempt	140	0.077	[0.035, 0.119]	−0.025	[-0.053, -0.000]
	anticipation	131	0.082	[0.027, 0.136]	−0.023	[-0.105, 0.050]

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Table 4 – Continued from previous page

Target	Label	Supp.	F1	95% CI	Skill	95% CI
	guilt	77	0.036	[0.000, 0.073]	−0.003	[-0.037, 0.029]
	trust	46	0.081	[0.012, 0.150]	+0.053	[-0.016, 0.120]
Gemma (PLarge)	longing	417	0.254	[0.228, 0.280]	−0.051	[-0.119, 0.002]
	anxiety	246	0.326	[0.237, 0.414]	+0.153	[0.057, 0.252]
	disgust	193	0.177	[0.080, 0.274]	+0.071	[-0.046, 0.184]
	hope	192	0.106	[0.051, 0.162]	+0.008	[-0.037, 0.054]
	shame	192	0.104	[0.044, 0.163]	+0.034	[0.001, 0.066]
	guilt	94	0.030	[0.002, 0.058]	−0.011	[-0.066, 0.038]
	confusion	89	0.012	[0.000, 0.027]	−0.008	[-0.018, 0.002]

4 Diagnostic checks

4.1 Centroid Distance per Annotation Scheme

Family	Condition	Mean	Min	Max	Closest Pair	Furthest Pair
Human	Annotator A	0.952	0.107	1.815	anger ↔ boldness	boldness ↔ sadness
	Annotator B	0.982	0.083	1.688	fear ↔ sadness	confidence ↔ despair/sadness
	AriEmozione 1	0.877	0.240	1.452	love ↔ sadness	admiration ↔ sadness
LLM-HTax	AnnA × Gemma (HT)	0.887	0.129	1.569	contempt/aversion ↔ sadness	boldness ↔ contempt/aversion
	AnnA × Phi4 (HT)	0.913	0.230	1.759	despair ↔ love	contempt/pity ↔ fear
	AnnB × Gemma (HT)	0.761	0.370	1.159	indignation/ sadness ↔ self-satisfaction	confidence ↔ sadness/ love
	AnnB × Phi4 (HT)	0.905	0.256	1.681	fear ↔ sadness	love/ sadness ↔ sadness
	AriEm1 × Gemma (HT)	0.854	0.232	1.375	anger ↔ fear	fear ↔ sadness
	AriEm1 × Phi4 (HT)	0.694	0.265	1.079	love ↔ sadness	fear ↔ love
LLM-PSmall	AnnA × Gemma (PS)	0.939	0.333	1.598	hope ↔ regret	hope ↔ trepidation
	AnnA × Phi4 (PS)	0.916	0.082	1.672	hope ↔ yearning	awe ↔ jealousy
	AnnB × Gemma (PS)	0.749	0.175	1.589	regret ↔ yearning	hope ↔ resentment
	AnnB × Phi4 (PS)	0.903	0.123	1.733	hope ↔ melancholy	awe ↔ hope
	AriEm1 × Gemma (PS)	0.912	0.268	1.636	guilt ↔ hope	guilt ↔ surprise
	AriEm1 × Phi4 (PS)	0.866	0.139	1.571	disgust ↔ trust	anticipation ↔ trust
LLM-Desc	Gemma (Desc)	0.914	0.316	1.582	astonishment ↔ fear	astonishment ↔ remorse
	Phi4 (Desc)	0.921	0.136	1.608	love/remorse ↔ love/sadness	boldness ↔ love/remorse
LLM-PLarge	Gemma (PLarge)	0.729	0.168	1.489	hope ↔ longing	anxiety ↔ longing
	Phi4 (PLarge)	0.863	0.397	1.714	anticipation ↔ despair	hope ↔ pride

Table 5: Summary of centroid distances by taxonomy and prompting family. Closest and furthest pairs are shown for each scheme.

4.2 Feature importance

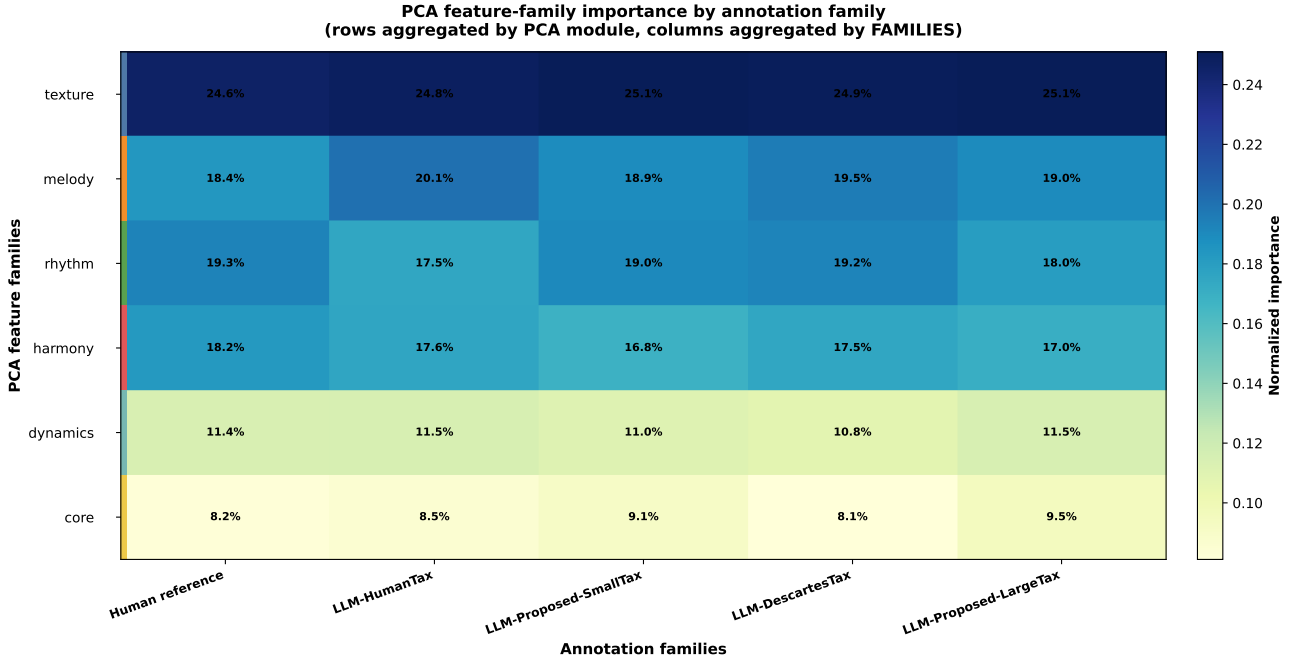


Figure 1: PCA feature-family importance by annotation family (rows aggregated by PCA module, columns aggregated by annotation schemes families)

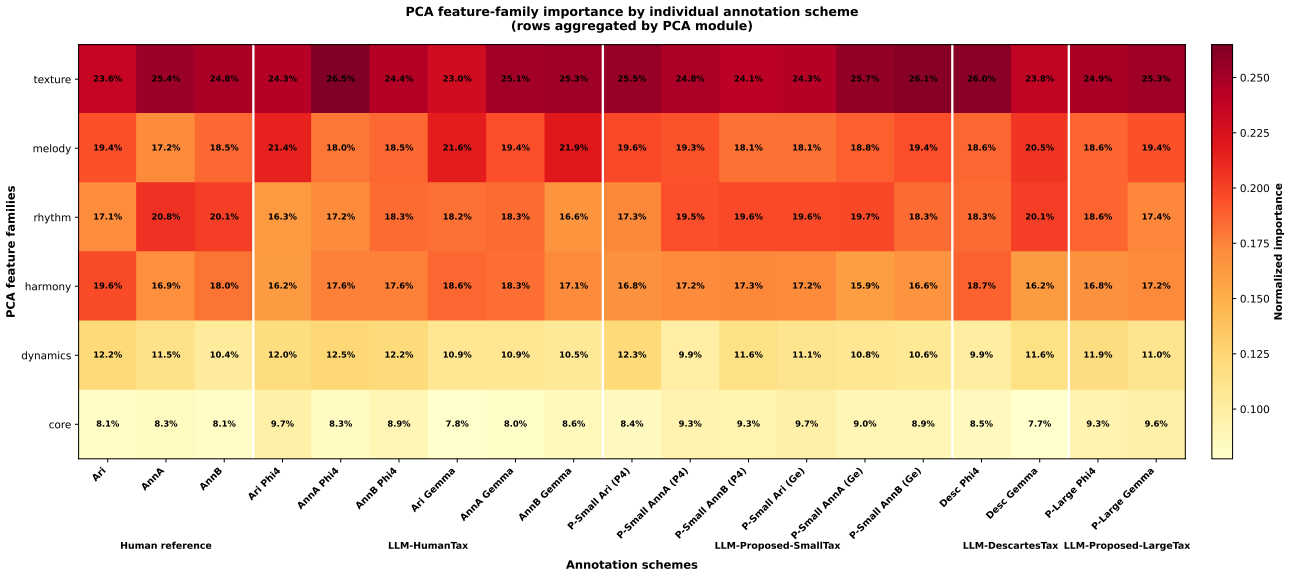


Figure 2: PCA feature-family importance by annotation scheme (rows aggregated by PCA module)

4.3 Robustness checks

Family	Condition	LogReg	SVM	RF	Best
Human	Annotator A	**0.024**	0.022	0.006	LogReg
	Annotator B	**0.036**	0.033	0.019	LogReg
	AriEmozione 1	**0.045**	0.020	-0.003	LogReg
LLM-HTax	AnnA × Phi4 (HT)	**0.019**	0.002	-0.009	LogReg
	AnnB × Phi4 (HT)	0.014	0.019	**0.027**	RF
	AriEm1 × Phi4 (HT)	**0.034**	-0.002	-0.026	LogReg
	AnnA × Gemma (HT)	**0.048**	0.034	-0.001	LogReg
	AnnB × Gemma (HT)	0.017	**0.018**	-0.012	SVM
	AriEm1 × Gemma (HT)	**0.112**	0.099	0.008	LogReg
LLM-PSmall	AnnA × Phi4 (PS)	0.024	**0.026**	**0.026**	SVM
	AnnB × Phi4 (PS)	**0.050**	0.046	0.026	LogReg
	AriEm1 × Phi4 (PS)	**0.017**	0.002	-0.021	LogReg
	AnnA × Gemma (PS)	**0.017**	0.012	0.010	LogReg
	AnnB × Gemma (PS)	**0.015**	0.012	0.001	LogReg
	AriEm1 × Gemma (PS)	**0.064**	0.028	0.013	LogReg
LLM-Desc	Phi4 (Desc)	**0.050**	0.042	-0.015	LogReg
	Gemma (Desc)	**0.008**	0.005	-0.023	LogReg
LLM-PLarge	Phi4 (PLarge)	**0.032**	0.025	0.009	LogReg
	Gemma (PLarge)	0.031	0.027	**0.032**	RF

Table 6: Model skill scores (robustness check without outliers). Bold values indicate the best performer for each condition.

Comparison	Spearman ρ	p-value
LogReg vs. SVM	0.764	0.0001
LogReg vs. RF	0.142	0.5615
SVM vs. RF	0.525	0.0211

Table 7: Spearman rank correlations between model rankings.

4.4 Pairwise Adjusted Rand Index

Table 8: Pairwise Adjusted Rand Index (ARI) between all annotation conditions. Only pairs sharing at least 10 aria texts are included. Sorted by ARI descending.

Scheme A	Scheme B	N arias	ARI	Agreement
AnnA×Gemma (PS)	AnnB×Gemma (PS)	57	0.5464	Moderate
AnnB×Phi4 (PS)	AnnA×Phi4 (HT)	52	0.3955	Weak
Gemma (Desc)	AriEm1×Gemma (HT)	81	0.3924	Weak
AnnA×Phi4 (PS)	AnnB×Phi4 (PS)	63	0.3871	Weak
Phi4 (PLarge)	AnnA×Phi4 (HT)	55	0.3834	Weak
AriEm1×Phi4 (HT)	AriEm1×Gemma (HT)	82	0.3695	Weak
AnnA×Phi4 (HT)	AnnA×Gemma (HT)	59	0.3636	Weak
Annotator A	Annotator B	39	0.3630	Weak
Gemma (PLarge)	AriEm1×Gemma (PS)	83	0.3613	Weak
AnnA×Phi4 (HT)	AriEm1×Phi4 (HT)	63	0.3479	Weak
AriEm1×Phi4 (HT)	Gemma (Desc)	80	0.3411	Weak
AnnB×Phi4 (HT)	AriEm1×Phi4 (HT)	61	0.3372	Weak
AnnA×Phi4 (HT)	AnnB×Phi4 (HT)	51	0.3230	Weak
Phi4 (PLarge)	AnnA×Gemma (PS)	62	0.3228	Weak
AnnA×Phi4 (HT)	AnnB×Gemma (PS)	46	0.3073	Weak
AriEmozione 1	Annotator A	61	0.3062	Weak
AnnA×Phi4 (PS)	Gemma (PLarge)	70	0.3057	Weak
Phi4 (PLarge)	AnnA×Gemma (HT)	67	0.3045	Weak
AnnA×Gemma (HT)	AriEm1×Gemma (HT)	75	0.2965	Weak
AriEm1×Phi4 (PS)	Phi4 (Desc)	75	0.2934	Weak
AnnA×Phi4 (PS)	AnnA×Phi4 (HT)	55	0.2866	Weak
Phi4 (PLarge)	AriEm1×Gemma (PS)	71	0.2829	Weak
Annotator A	AnnB×Phi4 (HT)	48	0.2776	Weak
AnnB×Phi4 (HT)	AnnA×Gemma (HT)	58	0.2767	Weak
Phi4 (PLarge)	AnnB×Phi4 (HT)	55	0.2731	Weak
Annotator A	AnnA×Phi4 (PS)	51	0.2716	Weak
AnnA×Phi4 (PS)	Phi4 (Desc)	64	0.2690	Weak
Gemma (Desc)	AnnA×Gemma (HT)	74	0.2665	Weak
AnnA×Phi4 (HT)	Gemma (Desc)	64	0.2626	Weak
Annotator B	AnnB×Phi4 (PS)	42	0.2623	Weak
Phi4 (PLarge)	AnnB×Gemma (HT)	49	0.2621	Weak
AriEmozione 1	AriEm1×Gemma (HT)	83	0.2586	Weak
Annotator B	AnnA×Phi4 (PS)	42	0.2583	Weak
Phi4 (PLarge)	AnnA×Phi4 (PS)	63	0.2582	Weak
Phi4 (PLarge)	Phi4 (Desc)	63	0.2578	Weak
AnnB×Phi4 (PS)	AnnB×Phi4 (HT)	54	0.2551	Weak
Phi4 (PLarge)	AnnB×Gemma (PS)	54	0.2548	Weak
AnnA×Phi4 (PS)	AriEm1×Gemma (PS)	70	0.2523	Weak
AnnB×Gemma (PS)	AriEm1×Gemma (PS)	60	0.2503	Weak
AnnB×Phi4 (PS)	Gemma (PLarge)	68	0.2444	Weak
Phi4 (PLarge)	AnnB×Phi4 (PS)	61	0.2416	Weak
Annotator A	AnnA×Gemma (HT)	56	0.2337	Weak
AnnB×Gemma (PS)	AnnA×Gemma (HT)	58	0.2326	Weak
AnnA×Gemma (PS)	AnnA×Gemma (HT)	67	0.2315	Weak
Annotator B	AnnA×Phi4 (HT)	38	0.2300	Weak
AnnA×Gemma (PS)	AnnB×Gemma (HT)	55	0.2294	Weak
AnnB×Phi4 (PS)	AnnA×Gemma (PS)	61	0.2292	Weak
AnnA×Gemma (PS)	AriEm1×Gemma (PS)	73	0.2286	Weak
AriEmozione 1	AriEm1×Phi4 (HT)	82	0.2286	Weak

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Table 8 – Continued from previous page

Scheme A	Scheme B	N arias	ARI	Agreement
AnnA×Gemma (HT)	AnnB×Gemma (HT)	57	0.2264	Weak
AnnB×Phi4 (HT)	Gemma (PLarge)	62	0.2258	Weak
AnnA×Phi4 (PS)	AnnB×Gemma (PS)	53	0.2233	Weak
AnnA×Phi4 (PS)	AnnB×Phi4 (HT)	55	0.2231	Weak
AnnB×Phi4 (PS)	AriEm1×Phi4 (PS)	69	0.2228	Weak
AriEm1×Phi4 (HT)	AnnA×Gemma (HT)	74	0.2220	Weak
Annotator A	Gemma (PLarge)	60	0.2215	Weak
AnnB×Phi4 (PS)	AnnA×Gemma (HT)	61	0.2209	Weak
Annotator B	AnnB×Gemma (PS)	36	0.2202	Weak
AnnA×Phi4 (HT)	AnnA×Gemma (PS)	57	0.2183	Weak
AriEmozione 1	AnnB×Gemma (PS)	60	0.2180	Weak
Annotator A	AnnB×Phi4 (PS)	51	0.2170	Weak
AnnA×Phi4 (PS)	AriEm1×Phi4 (PS)	70	0.2170	Weak
AnnA×Phi4 (PS)	AnnA×Gemma (PS)	61	0.2164	Weak
AnnB×Phi4 (HT)	Gemma (Desc)	61	0.2160	Weak
Phi4 (PLarge)	Gemma (Desc)	69	0.2147	Weak
AnnA×Gemma (PS)	AriEm1×Gemma (HT)	72	0.2134	Weak
AnnB×Phi4 (HT)	AriEm1×Gemma (HT)	62	0.2100	Weak
Phi4 (Desc)	AriEm1×Phi4 (HT)	73	0.2094	Weak
AriEm1×Phi4 (PS)	AriEm1×Gemma (PS)	84	0.2077	Weak
Phi4 (PLarge)	AriEm1×Phi4 (HT)	69	0.2058	Weak
AnnB×Phi4 (HT)	AnnA×Gemma (PS)	55	0.2050	Weak
Phi4 (PLarge)	AriEm1×Phi4 (PS)	71	0.2038	Weak
AnnA×Phi4 (PS)	AnnA×Gemma (HT)	65	0.2038	Weak
Annotator A	AnnB×Gemma (PS)	45	0.2031	Weak
Annotator A	AriEm1×Gemma (HT)	61	0.2017	Weak
AriEm1×Phi4 (PS)	AnnA×Phi4 (HT)	65	0.1995	Weak
AnnB×Phi4 (PS)	AnnB×Gemma (PS)	52	0.1993	Weak
AriEmozione 1	AnnA×Gemma (PS)	73	0.1991	Weak
AnnA×Phi4 (HT)	AnnB×Gemma (HT)	47	0.1988	Weak
AriEm1×Gemma (PS)	AnnA×Gemma (HT)	76	0.1983	Weak
AriEm1×Phi4 (PS)	AriEm1×Phi4 (HT)	82	0.1980	Weak
Annotator A	AnnA×Phi4 (HT)	48	0.1964	Weak
AnnB×Phi4 (PS)	AriEm1×Gemma (PS)	69	0.1953	Weak
AriEm1×Phi4 (PS)	AnnB×Phi4 (HT)	63	0.1926	Weak
Annotator B	AriEm1×Phi4 (PS)	50	0.1926	Weak
AnnA×Gemma (PS)	Gemma (Desc)	71	0.1921	Weak
Gemma (PLarge)	AnnA×Gemma (PS)	72	0.1920	Weak
Annotator B	AnnA×Gemma (HT)	45	0.1917	Weak
AnnB×Gemma (PS)	AnnB×Gemma (HT)	47	0.1915	Weak
AriEm1×Phi4 (HT)	AnnA×Gemma (PS)	71	0.1913	Weak
Gemma (PLarge)	AnnB×Gemma (PS)	60	0.1905	Weak
Phi4 (PLarge)	AriEm1×Gemma (HT)	70	0.1898	Weak
Annotator B	AnnA×Gemma (PS)	44	0.1891	Weak
AriEm1×Phi4 (HT)	AriEm1×Gemma (PS)	82	0.1872	Weak
AnnA×Phi4 (HT)	AriEm1×Gemma (PS)	65	0.1862	Weak
Phi4 (Desc)	AnnA×Phi4 (HT)	59	0.1859	Weak
AnnA×Phi4 (HT)	Gemma (PLarge)	64	0.1842	Weak
AnnB×Phi4 (HT)	AnnB×Gemma (PS)	48	0.1828	Weak
Phi4 (Desc)	AnnB×Phi4 (HT)	58	0.1822	Weak
Annotator B	AriEm1×Gemma (PS)	50	0.1807	Weak
Phi4 (Desc)	AnnB×Gemma (PS)	55	0.1785	Weak
AnnB×Phi4 (PS)	Gemma (Desc)	67	0.1782	Weak
Annotator B	AnnB×Phi4 (HT)	38	0.1781	Weak

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Scheme A	Scheme B	N arias	ARI	Agreement
Annotator A	AnnA×Gemma (PS)	54	0.1745	Weak
AnnB×Phi4 (HT)	AriEm1×Gemma (PS)	63	0.1726	Weak
AnnB×Gemma (PS)	AriEm1×Gemma (HT)	59	0.1721	Weak
AnnA×Phi4 (HT)	AriEm1×Gemma (HT)	64	0.1721	Weak
AriEmozione 1	Phi4 (Desc)	75	0.1720	Weak
AriEm1×Phi4 (HT)	AnnB×Gemma (PS)	59	0.1718	Weak
AriEmozione 1	Annotator B	50	0.1713	Weak
AriEmozione 1	AnnA×Phi4 (PS)	70	0.1705	Weak
Phi4 (Desc)	Gemma (Desc)	73	0.1694	Weak
AriEm1×Gemma (PS)	AriEm1×Gemma (HT)	83	0.1690	Weak
Phi4 (Desc)	AnnA×Gemma (HT)	69	0.1690	Weak
AnnB×Phi4 (PS)	Phi4 (Desc)	62	0.1678	Weak
Gemma (Desc)	AnnB×Gemma (HT)	60	0.1666	Weak
AnnA×Phi4 (PS)	AnnB×Gemma (HT)	51	0.1658	Weak
AnnB×Gemma (PS)	Gemma (Desc)	58	0.1655	Weak
Phi4 (Desc)	AnnA×Gemma (PS)	67	0.1652	Weak
Annotator A	AriEm1×Phi4 (HT)	60	0.1645	Weak
AriEmozione 1	Phi4 (PLarge)	71	0.1632	Weak
AriEmozione 1	Gemma (PLarge)	83	0.1613	Weak
AriEmozione 1	AriEm1×Gemma (PS)	84	0.1604	Weak
AnnB×Phi4 (PS)	AriEm1×Phi4 (HT)	68	0.1585	Weak
AriEm1×Phi4 (HT)	Gemma (PLarge)	81	0.1579	Weak
AnnA×Phi4 (PS)	AriEm1×Phi4 (HT)	69	0.1557	Weak
Annotator B	Phi4 (Desc)	43	0.1543	Weak
AriEm1×Phi4 (PS)	Gemma (PLarge)	83	0.1519	Weak
Phi4 (Desc)	AriEm1×Gemma (HT)	74	0.1499	Weak
Annotator A	Phi4 (PLarge)	53	0.1494	Weak
AriEmozione 1	AnnB×Phi4 (HT)	63	0.1483	Weak
Annotator A	Gemma (Desc)	59	0.1480	Weak
Annotator A	AriEm1×Gemma (PS)	61	0.1473	Weak
Annotator B	Gemma (PLarge)	49	0.1472	Weak
AriEm1×Phi4 (PS)	AnnB×Gemma (PS)	60	0.1466	Weak
AriEm1×Phi4 (PS)	AnnA×Gemma (PS)	73	0.1458	Weak
Gemma (PLarge)	AnnA×Gemma (HT)	76	0.1435	Weak
AriEmozione 1	AriEm1×Phi4 (PS)	84	0.1410	Weak
AriEm1×Phi4 (PS)	AnnA×Gemma (HT)	76	0.1401	Weak
AriEm1×Gemma (PS)	Gemma (Desc)	82	0.1380	Weak
AriEmozione 1	Gemma (Desc)	82	0.1368	Weak
Annotator A	AnnB×Gemma (HT)	44	0.1338	Weak
AnnB×Phi4 (PS)	AnnB×Gemma (HT)	50	0.1308	Weak
Annotator A	AriEm1×Phi4 (PS)	61	0.1302	Weak
Gemma (PLarge)	Gemma (Desc)	81	0.1302	Weak
Gemma (PLarge)	AriEm1×Gemma (HT)	82	0.1283	Weak
Phi4 (PLarge)	Gemma (PLarge)	71	0.1244	Weak
Annotator B	AnnB×Gemma (HT)	36	0.1237	Weak
AnnA×Phi4 (PS)	AriEm1×Gemma (HT)	69	0.1234	Weak
Phi4 (Desc)	AriEm1×Gemma (PS)	75	0.1230	Weak
Annotator B	AriEm1×Gemma (HT)	49	0.1216	Weak
Annotator A	Phi4 (Desc)	54	0.1185	Weak
AriEmozione 1	AnnA×Phi4 (HT)	65	0.1179	Weak
AriEmozione 1	AnnA×Gemma (HT)	76	0.1158	Weak
AriEm1×Phi4 (PS)	Gemma (Desc)	82	0.1133	Weak
Annotator B	Phi4 (PLarge)	43	0.1114	Weak
AriEm1×Gemma (PS)	AnnB×Gemma (HT)	61	0.1105	Weak

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Table 8 – Continued from previous page

Scheme A	Scheme B	N arias	ARI	Agreement
Annotator B	Gemma (Desc)	49	0.1091	Weak
AnnB×Phi4 (HT)	AnnB×Gemma (HT)	48	0.1091	Weak
AnnB×Phi4 (PS)	AriEm1×Gemma (HT)	68	0.1062	Weak
Phi4 (Desc)	Gemma (PLarge)	74	0.1050	Weak
AnnB×Gemma (HT)	AriEm1×Gemma (HT)	61	0.1036	Weak
AnnA×Phi4 (PS)	Gemma (Desc)	68	0.1019	Weak
Annotator B	AriEm1×Phi4 (HT)	49	0.1006	Weak
AriEmozione 1	AnnB×Phi4 (PS)	69	0.0999	Chance
AriEm1×Phi4 (HT)	AnnB×Gemma (HT)	60	0.0940	Chance
Phi4 (Desc)	AnnB×Gemma (HT)	57	0.0853	Chance
Gemma (PLarge)	AnnB×Gemma (HT)	61	0.0846	Chance
AriEm1×Phi4 (PS)	AnnB×Gemma (HT)	61	0.0707	Chance
AriEmozione 1	AnnB×Gemma (HT)	61	0.0706	Chance
AriEm1×Phi4 (PS)	AriEm1×Gemma (HT)	83	0.0645	Chance

5 Computational Setup and Reproducibility

All experiments were conducted using a local inference framework to ensure data privacy and the elimination of external API variability.

5.1 Hardware Resources

The experiments were executed on a **MacBook Air** powered by the **Apple M3 chip** (8-core CPU/GPU architecture). The system was equipped with **24 GB of unified memory**, which provided the necessary bandwidth and capacity to host the 12B and 14B parameter models locally without offloading.

5.2 Software and Model Specifications

Local model serving was managed via **Ollama**. For reproducibility, we utilize the specific model digests (IDs) to identify the exact binary weights used in this study:

- **Phi-4 (14.7B)**:
  - *Ollama ID*: `ac896e5b8b34`
  - *Architecture*: Phi-3/4 based transformer.
  - *Quantization*: 4-bit (Q4_K_M), resulting in a 9.1 GB disk footprint.
  - *Context Window*: 16,384 tokens.
- **Gemma 3 (12.2B)**:
  - *Ollama ID*: `f4031aab637d`
  - *Architecture*: Gemma 3.
  - *Quantization*: 4-bit (Q4_K_M), resulting in an 8.1 GB disk footprint.
  - *Context Window*: 131,072 tokens.

5.3 Inference Configuration

To ensure deterministic results for taxonomy mapping and distance calculations, the **temperature** was set to 0.0 for all production runs. Other parameters followed the model defaults (e.g., Top-K = 64, Top-P = 0.95 for Gemma 3).

5.4 Detailed Model Manifests

The following table summarizes the technical configuration of the local LLMs as reported by `ollama show -verbose` and `ollama list`.

Property	**Phi-4 (latest)**	**Gemma 3 (12b)**
Ollama ID (Digest)	`ac896e5b8b34`	`f4031aab637d`
File Size	9.1 GB	8.1 GB
Parameter Count	14.7B	12.2B
Architecture	phi3	gemma3
Context Length	16,384	131,072
Embedding Length	5,120	3,840
Quantization	Q4_K_M	Q4_K_M
Stop Sequence	`<|end_of_text|>`	`<end_of_turn>`

Table 9: Technical specifications for local model reproducibility.